Chapter 14 (New): The Technology Pillar — Data Infrastructure for a Transformed Health System

VHCURES, VITL, the Health Information Exchange, the Statewide EHR Question, FHIR Interoperability, AI Scribes, Remote Patient Monitoring, Telehealth, and the Clinically Integrated Network Vermont Is Building

Sources: GMCB VHCURES documentation; Vermont HIE Strategic Plan (2024 Update); Act 62 of 2025 (HIE governance transfer to DVHA); Oliver Wyman Act 167 Recommendations; AHS January 2025 Legislative Presentation; Vermont RHT Program Application and Activity Details (November–December 2025); Chartis 2026 State of Rural Health; JAMA Network Open ambient scribe study (2024); CMS FHIR Interoperability Rule; Vermont Blueprint for Health HIE data.

The Technology Pillar: Why Data Infrastructure Is the Foundation of Everything Else

There is a precise sense in which the Technology pillar is the precondition for every other pillar. Reference-based pricing requires knowing what each hospital actually charges for each service. Global budgets require attributing patients to hospitals and tracking total cost of care across the year. Clinical quality improvement requires measuring the gap between current performance and evidence-based targets. Health equity requires stratifying outcomes by race, income, and geography. Operational transformation requires modeling the financial and access impact of structural changes before implementing them.

All of these functions require data — specifically, data that is timely, complete, accurate, linked across clinical and claims sources, accessible to the organizations that need it, and governed in ways that protect patient privacy while enabling population-level analysis. Vermont's current data infrastructure — anchored by VHCURES (the all-payer claims database), VITL (the health information exchange), and the Blueprint clinical data registry — is sophisticated by national standards for a state of its size. It is also, in its current form, insufficient for the management demands of a system operating under global budgets and a mandatory Statewide Strategic Plan accountability framework.

This chapter develops Vermont's technology pillar comprehensively: what the current infrastructure does and does not do, where the critical gaps are, what the RHT Program is funding to close them, and what the national technology landscape — FHIR interoperability, ambient clinical intelligence, remote patient monitoring, AI diagnostic tools — means for Vermont's specific circumstances. It concludes with a technology architecture framework that Vermont's Strategic Plan should adopt and that any state-level transformation can use as a reference model.

Section 1: VHCURES — Vermont's All-Payer Claims Database

The Vermont Health Care Uniform Reporting and Evaluation System (VHCURES) is Vermont's all-payer claims database, managed by the Green Mountain Care Board with Onpoint Health Data as the data aggregation vendor. VHCURES has been operational in its current form since 2013 (when GMCB assumed responsibility from the Department of Financial Regulation) and is the primary analytical foundation for Vermont's healthcare regulatory functions.

What VHCURES Contains and What It Can Do

VHCURES is a substantial data asset that gives Vermont analytical capabilities most states lack. Understanding both its capabilities and its limitations is essential for anyone designing Vermont's technology infrastructure.

Commercial population covered	Government payer coverage
60% *Of commercially insured Vermonters — partial due to ERISA self-insured employer exemption*	**100%** *Of Medicaid and Medicare enrollees in Vermont*
Data types **Claims + Rx** *Medical and pharmacy claims, plus eligibility and demographic data*	**Years available per extract** **Up to 5** *Eligibility and paid claims; data lag typically 12-18+ months*

Figure 14.1 — VHCURES data coverage summary. Source: GMCB VHCURES documentation.

VHCURES enables Vermont to conduct analyses that directly support the transformation agenda: population-level total cost of care measurement (the GMCB publishes all-payer TCOC dashboards derived from VHCURES); potentially avoidable utilization analysis (the 32.3% avoidable ED visit rate in AHS's transformation reports is calculated from VHCURES via VUHDDS); hospital price transparency analysis (the GMCB's 2026 RBP report used VHCURES-derived 2024 claims data to show hospital prices ranging from 279% to 697% of Medicare for outpatient services); and prescription drug affordability analysis (GMCB's 2025 report on drug pricing used VHCURES to model the impact of IRA drug negotiations on Vermont spending). This is a powerful analytical platform that has produced some of the most credible and consequential health policy analysis in recent Vermont history.

VHCURES Limitations — Where the Gaps Are

VHCURES has four structural limitations that matter directly for Vermont's transformation management needs. Oliver Wyman identified the data infrastructure as one of the barriers to coordinated transformation, and the specific limitations below explain why.

Limitation	What it means for transformation	Required improvement
Incomplete commercial coverage (60%)	The 40% of commercially insured Vermonters in ERISA self-insured employer plans — many of them higher-income, higher-resource patients — are not in VHCURES unless their employer voluntarily submits. Total cost of care calculations and population health analyses are systematically missing a significant portion of the commercially insured population. Under federal law, self-insured employers cannot be required to submit data to state APCDs.	Voluntary submission incentives; federal ERISA reform advocacy; use of commercial insurer aggregate data for the covered portion; flag analyses as partial in all public reporting
Data lag of 12-18+ months	The most recent VHCURES data typically reflects conditions 12-18 months prior to analysis. The November 2025 AHS transformation report explicitly acknowledged that most available measures reflect 2023 or earlier data. Under global budgets, where hospitals need to understand their population health performance in near-real-time to manage within their budget cap, this lag makes VHCURES insufficient as a management tool.	AHEAD Model requires more timely data reporting; RHT Program analytics investments; GMCB FY27 hospital budget guidance requires hospitals to report more frequently; push for 60-90 day claims lag instead of 12-18 months

No behavioral health or SDOH data integration	VHCURES contains medical and pharmacy claims but not mental health and SUD clinical records (governed by 42 CFR Part 2 and state confidentiality laws), social service utilization data, housing status, or other SDOH indicators. The HIE Steering Committee has a statutory directive to create one health record per person integrating claims, clinical, and SDOH data — but this integration has not been implemented.	VHCURES-VHIE integration per HIE Steering Committee statutory mandate; Act 167 HIE provisions; SDOH data collection through Blueprint HRSN screening linked to VHCURES identifiers with appropriate consent framework
Voluntary provider participation in HIE	VITL's VHIE receives clinical data from hospitals but participation by community-based providers, mental health agencies, skilled nursing facilities, and other post-acute providers is voluntary and incomplete. Oliver Wyman specifically cited VITL as lacking pharmacy claims data and not being viewed as user-friendly. Act 68 requires hospitals to maintain HIE connectivity but does not mandate all provider types.	Act 68 hospital HIE connectivity requirement is a start; AHEAD requires primary care participation; RHT Program telehealth investment includes interoperability requirements; progressive connectivity incentives for non-hospital providers

Figure 14.2 — VHCURES structural limitations and required improvements. Sources: GMCB VHCURES data documentation; Oliver Wyman Act 167 Report; AHS Transformation Reports; Act 68 of 2025.

VHCURES Perception vs. Reality — Oliver Wyman's Finding

Oliver Wyman's perception-versus-reality analysis identified Vermont's data infrastructure as one of the areas where the gap between apparent capability and actual capability is most consequential. The relevant finding:

Perception: Vermont has a State Health Information Exchange (VITL).

Reality: Participation in VITL is voluntary and many community-based providers are not included. VITL lacks pharmacy claims data and is not viewed as 'user friendly' by providers.

This matters because Vermont's transformation strategy depends on care coordination between hospitals and community providers — and care coordination requires data sharing. A care coordinator trying to manage a patient transitioning from a hospital inpatient stay to a community mental health program cannot access the clinical data they need if the mental health agency is not connected to VITL. A primary care practice trying to close chronic disease management gaps cannot see the specialty care their patient received at a different facility if that facility's data is not in the HIE.

Vermont's technology investments — the VHCURES-VHIE integration mandate, the RHT Program interoperability grants, the AHEAD data sharing requirements — are all aimed at closing this gap between VITL's nominal existence and the functional interoperability Vermont's care coordination model requires.

Section 2: The Vermont HIE — VITL, the Governance Shift, and the Integration Imperative

Vermont's Health Information Exchange (VHIE) is operated by Vermont Information Technology Leaders (VITL), an independent nonprofit designated by statute as the exclusive operator of the statewide HIE network. Every hospital in Vermont contributes records to the VHIE. The VHIE operates a provider portal (VITLAccess) and a query service that allows providers to access patient records from other facilities. Vermont has been building this infrastructure since 2009, when VITL was designated as the Regional Extension Center to help primary care providers adopt electronic health records.

The 2025 Governance Shift: Act 62 Transfers HIE Authority to DVHA

Effective July 1, 2025, Act 62 of 2025 made a significant governance change: responsibility for Vermont's statewide Health Information Technology Plan was transferred from the Green Mountain Care Board to the Department of Vermont Health Access (DVHA). Prior to this, GMCB had authority to review and approve the HIE Strategic Plan, VHIE Connectivity Criteria, and VITL's budget. Under Act 62, DVHA now coordinates Vermont's statewide HIT Plan in consultation with the HIE Steering Committee, with annual revisions submitted by November 1 and comprehensive updates every five years.

The governance rationale for this shift is significant: DVHA has direct operational relationships with healthcare providers through its Medicaid contracts, Blueprint for Health administration, and AHEAD Model management. Moving HIT plan responsibility to DVHA aligns the technology strategy with the healthcare delivery transformation authority. It also means that Vermont's AHEAD implementation — which requires robust data exchange for population health management and global budget monitoring — is now coordinated by the same agency managing the data infrastructure.

The Integration Mandate: One Health Record Per Person

The Vermont legislature gave the HIE Steering Committee an explicit statutory mandate that defines the technology pillar's central ambition for the next five years: create one health record for each person that integrates claims data, clinical data, mental health and substance use disorder services data, and social determinants of health data. This mandate is documented in state law and in AHS's own letters to the legislature. It is the most ambitious data integration goal any state has formally committed to, and its achievement would give Vermont a population health management capability that no other state currently possesses.

The practical path to this integration has three components that must be executed in sequence:

Step	What it requires	Timeline	Current status
1. VHCURES-VHIE claims-clinical integration	Merge VHCURES claims data with VHIE clinical data using a privacy-compliant patient matching methodology. This gives every provider record access to both the claims history (what care was delivered, at what cost, through what payers) and the clinical record (diagnoses, medications, lab results, care plans).	DVHA leading; AHEAD Model requires this for population health management; target 2026-2027	HIE Steering Committee has a statutory directive to include data integration strategy in HIE plans; VHCURES-VITL linkage already used for Blueprint HEDIS measurement; full bidirectional integration not yet operational
2. Mental health and SUD data integration	Add mental health and substance use disorder data to the integrated record, with appropriate consent frameworks that comply with 42 CFR Part 2 (federal SUD confidentiality rules) and	2027-2028; requires federal 42 CFR Part 2 compliance pathway	DVHA working with VITL on consent policy; 2025 federal regulatory changes to 42 CFR Part 2 may simplify compliance pathway; behavioral health CCBHC expansion creates urgency

	Vermont's own privacy statutes. This is technically complex — SUD data has a separate consent regime from medical data — but clinically essential.		
3. SDOH data integration	Add social service utilization data — housing program participation, food assistance, transportation use, domestic violence services — linked to the health record with appropriate privacy protections. This is the most politically sensitive and technically complex integration because it involves non-healthcare agencies and data that carries significant stigma risk.	2028 and beyond; requires cross-agency data governance agreement	Blueprint HRSN screening generates SDOH data but it is not yet linked to VHCURES or VHIE at scale; AHS HSA Coordinator model (Chapter 11) provides the cross-agency governance vehicle

Figure 14.3 — Vermont's three-step path to integrated health records. Sources: Vermont HIE Steering Committee statutory mandate; AHS January 2025 Legislative Presentation; DVHA HIE governance documents.

The Statewide EHR Question — Vermont's Active Feasibility Assessment

Oliver Wyman's community meetings generated a clear finding about EHR fragmentation: providers cannot efficiently coordinate care across facilities when different hospitals run different EHR systems that do not share data effectively. Vermont's 14 hospitals currently use four different EHR vendors — Epic (dominant at UVMMC), Oracle Health (formerly Cerner), TruBridge (formerly CPSI, used by smaller community hospitals), and Meditech — plus various additional software for laboratory management, pharmacy, and other functions.

AHS has commissioned a feasibility assessment to evaluate the cost-benefit of implementing a statewide electronic health record — a single EHR platform that all Vermont hospitals and primary care practices would use, enabling seamless data exchange as a byproduct of shared infrastructure rather than through complex interface work. Vermont's HIE Strategic Plan describes the assessment as evaluating 'what a statewide EHR would look like in line with the Act 167 Report.'

Case for a statewide EHR	Case against (or caution)
Eliminates interface complexity — data is naturally shared across providers on the same platform	Capital cost is substantial — major EHR implementations run $100-300M for large health systems; statewide scope amplifies this
Community meeting participants specifically cited EMR fragmentation as a barrier: 'EMR systems don't talk to each other and it's difficult to transfer from one center to another'	Disruption risk — forced EHR migrations have caused significant clinical workflow disruption and temporary quality problems at major health systems
Reduces per-provider IT infrastructure cost — smaller hospitals cannot afford full IT departments; shared infrastructure is more efficient	FHIR-based interoperability (see Section 4) may make data exchange manageable without a single EHR, reducing the case for the disruption of migration

Enables real-time population health data that VHCURES's 12-18 month lag cannot provide	Vendor lock-in — a single-vendor statewide EHR creates significant dependence on that vendor's pricing, contract terms, and long-term viability
Maryland's experience: common data infrastructure enabled the HSCRC to manage global budgets effectively	ERISA self-insured employers — outside state authority — would still not be in the system even with a statewide EHR
RHT Program IT advance funds can support the capital investment — this is an explicitly authorized use of RHT funds	Vermont's diverse provider landscape (hospitals, FQHCs, private practices, mental health agencies) may not all be able to adopt a single platform

Figure 14.4 — Vermont statewide EHR: case for and against. The feasibility assessment underway will weigh these factors against Vermont's specific cost, disruption tolerance, and FHIR interoperability capacity.

The feasibility assessment's most important analytical question is whether FHIR-based interoperability — the national standard for health data exchange that is rapidly maturing — can deliver the data sharing Vermont needs without the cost and disruption of a statewide EHR migration. If FHIR can deliver real-time, bi-directional data exchange across Vermont's four EHR platforms at an acceptable implementation cost, the case for a single statewide EHR weakens significantly. Section 4 develops the FHIR picture. AHS's feasibility assessment should produce a clear answer on this question by 2026-2027 — in time to inform the 2028 Strategic Plan.

Section 3: The Vermont Clinically Integrated Network — A New Technology Institution

Vermont's Rural Health Transformation Program application includes one of the most significant technology investments in the entire $195M award: the creation of a Clinically Integrated Network (CIN) linking Vermont's hospitals and community providers into a coordinated clinical and data-sharing infrastructure. Vermont is one of four states (along with Arkansas, Nevada, and Oregon) that explicitly included CIN creation in their RHT applications, reflecting a national trend toward CINs as the organizational vehicle for rural health system integration.

What a Clinically Integrated Network Does

A CIN is a formal network of healthcare providers — hospitals, primary care practices, specialists, post-acute providers — that share clinical protocols, performance data, and care coordination infrastructure while maintaining independent ownership and governance. CINs occupy the space between full health system merger (which requires consolidation of ownership) and loose referral relationships (which have no formal data sharing or accountability). For Vermont's 14 independent, non-profit hospitals — which cannot and should not be merged into a single system — a CIN provides the coordination benefits of integration without the legal and political costs of ownership consolidation.

What the Vermont CIN Will Provide

Vermont's RHT Program application describes the Shared Services initiative — which includes the CIN infrastructure — as providing five specific functions for participating hospitals and providers:

- Analytics support: A shared analytics platform available to all hospitals and non-hospital providers for transformation and regionalization planning — identifying areas most in need, modeling the impact of service line changes, and tracking quality and cost metrics. This is the analytics vendor that AHS and GMCB are jointly procuring.

- Shared clinical protocols: Standardized clinical pathways for high-volume conditions — sepsis, heart failure, COPD, mental health crisis — that all participating providers adopt, enabling consistent quality regardless of which facility a patient uses.
- Group purchasing: Coordinated supply chain and group purchasing for medical supplies, equipment, and technology — capturing the scale economies that individual small hospitals cannot achieve independently. Oliver Wyman identified this as one of the highest-leverage near-term cost reduction opportunities.
- Administrative shared services: Shared infrastructure for billing, coding, credentialing, HR, and IT support — reducing the administrative overhead that drives Vermont's $1,303 per-discharge administrative cost premium over national benchmarks.
- Referral network formalization: Structured referral agreements between RSC hospitals and Tier 2/3 facilities for each COE specialty — ensuring that the regionalization blueprint operates as a reliable clinical network rather than a set of informal relationships.

The CIN also serves as the governance vehicle for the technology investments Vermont's transformation requires. Individual small hospitals cannot justify the capital investment in AI scribe technology, remote patient monitoring equipment, telehealth infrastructure, or advanced analytics platforms. A CIN can negotiate enterprise-wide technology contracts that make these investments affordable for the entire network, deploy shared technology infrastructure that individual providers access as a service, and establish common data standards that make the integrated health record vision achievable.

Section 4: FHIR Interoperability — The National Standard and Vermont's Compliance Path

FHIR (Fast Healthcare Interoperability Resources) is the HL7 standard for healthcare data exchange that CMS mandated for payers and providers beginning in 2021 and 2022. FHIR defines a common data format and API structure that allows different health IT systems to exchange information without custom interfaces. It is the closest thing to a universal language for healthcare data that the industry has adopted, and its maturation over the past five years has fundamentally changed what is technically possible in health data exchange.

What FHIR Enables That Vermont Currently Cannot Do

Vermont's current health data infrastructure predates widespread FHIR adoption and was built around older data exchange standards. The operational implications are concrete:

- Patient matching across facilities: Without FHIR-based master patient index integration, Vermont still lacks reliable patient matching across all providers. Community meeting participants described needing to contact 31 separate EMS agencies for patient transfers — a fragmentation problem that a FHIR-enabled patient matching system would substantially reduce.
- Real-time care gap identification: FHIR-enabled EHR systems can query a patient's record at the point of care, identifying care gaps (missing preventive services, chronic disease management gaps) in real time during the clinical encounter. Vermont's current Blueprint data profile process delivers care gap information to practices, but with significant lag. FHIR enables the same information at the point of care.
- Attribution accuracy for global budgets: AHEAD's hospital global budgets require accurate attribution of patients to hospitals for Medicare FFS. FHIR-based data exchange enables more precise attribution by tracking care delivery across providers in near-real-time, reducing the attribution disputes that complicate global budget performance measurement.
- Population health management at scale: Vermont's Blueprint CHTs currently manage populations using a combination of claims data, clinical registry data, and provider-reported information. FHIR-based population health platforms can aggregate this data automatically, maintaining continuous patient registries that trigger care coordinator outreach when patients are overdue for care or when clinical indicators suggest increased risk.

Vermont's FHIR Compliance Status

Vermont's larger hospitals — particularly UVMMC with its Epic EHR — are FHIR-compliant for the CMS-mandated payer-to-patient and payer-to-payer API requirements. The critical gap is at smaller community hospitals (on TruBridge and Meditech platforms) and at community-based providers (mental health agencies, skilled nursing facilities, home health agencies) that lack FHIR-capable systems entirely.

Vermont's RHT Program IT advance investments address this gap directly. The program includes grants for technology upgrades at smaller and more rural facilities that cannot afford FHIR-capable EHR migrations independently. The CIN infrastructure also supports FHIR compliance by providing shared technical assistance, vendor negotiations for favorable pricing on FHIR-capable platforms, and shared technical staff who can support implementation at facilities that lack internal IT capacity.

FHIR compliance level	Vermont providers and current status
Full FHIR R4 compliance	UVMMC (Epic); Vermont hospitals on Oracle Health (Cerner) post-2022 upgrade; major commercial payers (BCBS VT, MVP)
Partial FHIR implementation	Central Vermont Medical Center; hospitals on Meditech Expanse (FHIR-capable but may require configuration)

Limited / no FHIR	Smaller community hospitals on TruBridge legacy; most Designated Mental Health Agencies; many skilled nursing facilities; independent primary care practices on older EHR versions
FHIR investment planned (RHT Program)	Rural and independent practices; smaller CAHs; targeted technical assistance through Vermont CIN shared services

Figure 14.5 — Vermont FHIR compliance landscape (estimated as of 2026). Sources: AHS Legislative Presentation January 2025; Vermont RHT Program Application; GMCB HIE connectivity criteria.

Section 5: AI and Digital Health Technologies — Vermont's Targeted Investment Strategy

Vermont's RHT Program application is notable among the 47 state applications submitted in late 2025 for its specificity about technology investments. Rather than generic technology grants, Vermont's technology investment strategy is structured around five specific technology categories, each directly linked to an identified gap in Vermont's current system. This is the correct approach to one-time capital investment in technology: identify the specific operational problems, fund the specific technologies that solve them, and measure whether the problems are resolved.

AI Scribe Technology: Addressing the Documentation Burden Crisis

Vermont's RHT Program application explicitly includes grants for providers to purchase AI scribe technology, noting that smaller and more rural and independent practices are struggling with access to the same technology that larger practices can afford. The application cites that AI scribes automate an average of over two hours of daily administrative work by transcribing and processing clinician-patient conversations.

This is a direct response to a Vermont workforce problem: physician and clinician burnout driven by documentation burden is a primary contributor to provider attrition and is reducing Vermont's effective primary care capacity. A provider spending two hours daily on documentation has approximately 25% less time for patient care than one whose documentation is automated. At Vermont's scale — where the primary care FTE shortage is already projected at 370 by 2030 — recovering two hours per provider per day through AI documentation support is equivalent to adding significant additional clinical capacity without hiring new providers.

The evidence base for AI scribe technology is strong and growing. A 2024 quality improvement study published in JAMA Network Open found that ambient scribe tools were associated with improved documentation efficiency and reduced documentation burden. Clinicians used the technology spent less time writing notes during visits and reported less after-hours charting. Community providers at national rural health conferences have independently identified ambient AI as the single technology investment most likely to address workforce shortages while supporting billing processes and clinical decision-making simultaneously.

Remote Patient Monitoring: Managing Chronic Disease Outside the Hospital

Vermont's RHT Program application includes grants for remote patient monitoring (RPM) equipment for use in home health, primary care/specialty, and community paramedicine settings. RPM enables earlier interventions and ongoing monitoring for chronic conditions — congestive heart failure, COPD, diabetes, hypertension — in the home to prevent hospital admissions or readmissions.

The economics of RPM under global budgets are compelling. Under fee-for-service, RPM generates separate billable encounters and competes with in-person visits for reimbursement. Under global budgets, RPM is a cost-reduction tool: every hospital admission or readmission prevented by early RPM-triggered intervention saves revenue under the fixed budget. Vermont's aging population — with high rates of CHF, COPD, and diabetes among the 65+ cohort — is precisely the population where RPM produces the largest clinical and financial benefit.

Vermont's rural geography amplifies RPM's value. A patient in the Northeast Kingdom managing heart failure who needs twice-weekly monitoring faces a two-hour round trip to a clinic. RPM provides the same monitoring without the trip — better for the patient (who avoids travel costs and time), better for the provider (who can monitor more patients with existing staff), and better for the system (which reduces transportation-related access barriers that drive missed appointments and deteriorating chronic disease management).

Telehealth: Specialty Access for Rural Vermont

Vermont's RHT Program application includes specific telehealth investments to expand specialty care access in rural hospitals — allowing specialists at RSC facilities to provide consultation to patients at Tier 2 and Tier 3 facilities without requiring patient transfer. This is the technology enabler for Oliver Wyman's regionalization blueprint: rather than requiring every COE specialty to maintain physical presence at every community hospital, telehealth enables COE specialists to provide consultation across the network while patients remain near their homes.

Vermont's broadband landscape is a constraint: the RHT application notes that 80% of rural Vermonters have broadband access, compared to 85% in non-rural areas. The gap, while relatively small by national rural standards, means that home-based telehealth cannot reach all Vermont patients. Vermont's proposed broadband expansion (Oliver Wyman's third priority legislative change recommendation) is the infrastructure prerequisite for telehealth reaching the communities that need it most. The RHT Program's explicit inclusion of broadband infrastructure as an allowable use reflects this reality.

Vermont's telehealth strategy from the RHT application has four specific applications: specialty consultations (neurology, psychiatry, oncology) at rural hospitals; tele-ICU support for rural hospitals managing critical patients before transfer; tele-behavioral health for community mental health and CCBHC services; and community paramedicine — enabling EMTs and paramedics to provide extended care in patients' homes with real-time physician supervision via telehealth, reducing 911 calls and ED visits for non-emergency conditions.

Diagnostic AI: The Emerging Standard of Care

The book's Part V discussion of diagnostic AI maturation — FDA-cleared AI diagnostic devices for radiology, pathology, ophthalmology, and cardiology becoming standard of care in high-volume settings — applies directly to Vermont's context with one important distinction: Vermont's smaller rural hospitals may be the settings that benefit most from diagnostic AI, precisely because they lack the specialist volume to maintain expert human diagnostic capacity in every specialty.

A rural Vermont hospital with one radiologist who handles all imaging across multiple modalities cannot maintain the same expert pattern recognition as a high-volume academic center reading 50 chest CTs daily. AI-assisted radiology reading — which has demonstrated accuracy equal to or exceeding specialist readers for specific modalities like chest X-ray, mammography, and retinal imaging — can partially close this quality gap without requiring additional specialist recruitment. This is not a replacement for radiologists; it is a tool that makes existing radiologists more accurate and more efficient, particularly in the high-volume, lower-complexity reads that constitute the majority of rural hospital imaging.

Vermont's RHT IT advance funding can support diagnostic AI implementation at smaller hospitals that cannot fund these investments independently. The CIN's shared service model is an appropriate delivery mechanism: a network-wide diagnostic AI contract that all participating hospitals access, with shared quality monitoring and governance, would provide far better economics and oversight than 14 individual hospital technology decisions.

Price Transparency Technology

Vermont's RHT Program application includes a price transparency and insurance competition initiative — building the technology infrastructure to make hospital pricing legible to patients, payers, and regulators. GMCB was already developing a hospital price transparency dashboard as of February 2026 (first release forthcoming), providing the public with access to 2024 claims data showing hospital prices relative to Medicare. The RHT investment extends this infrastructure to enable patient-facing price comparison tools that allow Vermonters to see what a procedure costs at different facilities before choosing where to receive care.

Price transparency technology is a complement to reference-based pricing, not a substitute for it. RBP directly limits what hospitals can charge; price transparency helps patients and payers identify and act on pricing differences within the RBP framework. The combination — mandatory price caps plus transparent pricing information — creates the closest approximation to a functioning market for hospital services that any regulatory framework has achieved.

Section 6: AI Governance — Vermont's Responsibility Before the Risk Arrives

The book's Part V discussion of AI governance as a clinical requirement — moving from a quality initiative to a regulatory requirement as FDA, ONC, and CMS develop AI governance frameworks — applies to Vermont with specific urgency. Vermont is making substantial AI investments through the RHT Program: AI scribe grants, diagnostic AI potential, RPM algorithms. Each of these involves clinical AI that has the potential to produce biased outputs, performance degradation over time, or harms to specific patient populations if not properly monitored.

Vermont does not currently have a statewide AI governance framework for clinical AI. Individual hospital systems have varying levels of AI governance maturity — UVMMC, with its Epic deployment and academic medical center sophistication, is likely more advanced than a smaller community hospital purchasing AI scribe software for the first time. The CIN infrastructure Vermont is building through the RHT Program creates the opportunity to establish a shared AI governance framework that applies consistently across all Vermont providers.

Vermont AI Governance Framework: Minimum Requirements

Drawing on the HTR AI Governance Checklist and Vermont's specific clinical AI deployment context, Vermont's Statewide Strategic Plan should establish minimum AI governance requirements for clinical AI deployed in Vermont healthcare settings. At minimum, these should include:

- Pre-deployment bias testing: Any AI model used for clinical decision support, diagnostic assistance, or care gap identification must be tested for differential performance across race/ethnicity, age, sex, income, and geographic subgroups before deployment. Vermont's equity goals require that AI tools do not introduce or amplify health disparities.
- Performance monitoring: Deployed AI tools must have ongoing performance monitoring with predefined thresholds that trigger review or suspension. An AI model trained on national data may perform differently in Vermont's specific patient population — particularly in rural areas and among dual-eligible patients with complex social needs.
- Transparency requirements: Patients must be informed when AI tools are involved in their care. Clinicians must understand the basis and limitations of AI outputs. 'Black box' AI with no explainability is inappropriate for clinical settings regardless of its overall accuracy.
- Vendor accountability: Technology vendors providing clinical AI to Vermont providers must agree to performance monitoring, bias testing disclosure, and incident reporting. Contract terms must include performance standards and remedies for underperformance.
- Equity monitoring: AI performance must be specifically monitored for Vermont's highest-risk equity populations — Northeast Kingdom rural residents, Medicaid patients, elderly patients with complex needs, and patients whose primary language is not English.

Vermont's RHT Program AI scribe grants should include AI governance requirements as a condition of funding. The CIN's shared services function is the natural organizational home for a shared AI governance committee that monitors all AI deployments across Vermont's hospital network.

Section 7: Vermont's Target Technology Architecture — A Framework for the Strategic Plan

Vermont's Statewide Health Care Delivery Strategic Plan must specify a technology architecture — the integrated set of data systems, exchange standards, governance structures, and analytical platforms that enable the transformed delivery system to be managed, monitored, and improved. The following architecture organizes Vermont's technology investments across five layers, from foundational infrastructure to advanced analytical applications.

Layer 5 (Top) — Advanced Analytics and AI Applications

- Population health analytics platform (joint AHS-GMCB analytics vendor): Models transformation scenarios, evaluates hospital proposals, tracks outcome metrics
- AI scribe and documentation automation: RHT Program grants to rural and independent practices; CIN enterprise contract
- Diagnostic AI: Network-wide deployment through CIN shared services with shared AI governance
- Remote patient monitoring: RPM for CHF, COPD, diabetes management; community paramedicine integration
- Price transparency dashboard: GMCB public tool; patient-facing comparison tools through RHT investment

Layer 4 — Population Health Management

- Blueprint clinical data registry: HEDIS measurement, care gap reporting, population risk stratification
- Attribution management: Patient-to-hospital attribution for AHEAD global budgets; primary care attribution for PC AHEAD
- SDOH screening data: Blueprint HRSN screening results linked to health records
- Care coordination registry: CHT patient registries; CCBHC care management data

Layer 3 — Integrated Health Record (Target State)

- VHCURES-VHIE integration: Claims data merged with clinical data per HIE Steering Committee mandate
- Mental health and SUD data integration: 42 CFR Part 2 compliant consent framework; CCBHC and DA data
- SDOH data integration: Social service utilization linked to health record with privacy protections
- Single patient identity: Master patient index across all 14 hospitals and community providers

Layer 2 — Data Exchange Infrastructure

- Vermont Health Information Exchange (VHIE/VITL): Clinical data exchange; hospital connectivity mandated by Act 68
- FHIR R4 APIs: CMS-mandated payer and provider APIs; expanding to community providers via RHT grants
- Vermont Clinically Integrated Network: Shared data standards, protocols, and exchange agreements across 14 hospitals
- CommonWell / Carequality: National network connectivity for out-of-state care records

Layer 1 (Foundation) — Core Data Assets

- VHCURES (all-payer claims database): 100% Medicaid/Medicare; 60% commercial; GMCB managed; Onpoint Health Data vendor

- VUHDDS (hospital discharge database): All 14 Vermont hospitals; all inpatient, outpatient, ED discharges
- EHR systems at 14 hospitals: Epic (UVMMC), Oracle Health (CVMC, others), TruBridge (smaller CAHs), Meditech (others)
- Blueprint clinical registry: Quality measures data linked to VHCURES for population-level HEDIS
- GMCB regulatory data: Hospital budgets, insurance rate filings, CON decisions, ACO budget data

Figure 14.6 — Vermont target technology architecture: five-layer model. Sources: GMCB VHCURES documentation; Vermont HIE Strategic Plan; Oliver Wyman Act 167 Report; Vermont RHT Program Application; Act 62 and Act 68 of 2025.

Two observations about this architecture are important for Vermont's technology planning. First, layers 3 and above represent target states, not current states. Vermont's current technology infrastructure is solid at Layers 1 and 2, developing at Layer 4, and nascent at Layers 3 and 5. The RHT Program, AHEAD requirements, and the AHS-GMCB analytics vendor procurement are all investments in accelerating the build-out from Layer 1-2 completeness toward Layer 3-5 capability. The Statewide Strategic Plan's technology section should be explicit about which layers are complete, which are under development, and what the completion timeline is.

Second, the technology architecture has a governance counterpart that is equally important. Each layer requires governance: who owns the data, who can access it under what conditions, what privacy protections apply, how disputes are resolved, and how the infrastructure is funded sustainably. Vermont's technology investments will only produce their intended value if the governance layer — the policies, agreements, and oversight structures — keeps pace with the technical infrastructure. The DVHA-led HIE governance, the GMCB's data stewardship program, and the CIN's shared governance model are all components of this governance architecture.

Key Concepts Introduced or Expanded in This Chapter

Term	Definition
VHCURES	Vermont Health Care Uniform Reporting and Evaluation System — Vermont's all-payer claims database, managed by GMCB with Onpoint Health Data. Contains 100% of Medicaid and Medicare claims and approximately 60% of commercial claims; used for total cost of care measurement, hospital price analysis, and population health analytics.
VITL / VHIE	Vermont Information Technology Leaders / Vermont Health Information Exchange — the nonprofit designated by statute to operate Vermont's statewide HIE network, providing clinical data exchange between hospitals and other providers. Participation is voluntary for non-hospital providers.
FHIR (Fast Healthcare Interoperability Resources)	The HL7 standard for healthcare data exchange, mandated by CMS for payers and providers beginning 2021-2022. FHIR enables different health IT systems to exchange information using common data formats and APIs, without custom interfaces.
Clinically Integrated Network (CIN)	A formal network of healthcare providers that share clinical protocols, performance data, and care coordination infrastructure while maintaining independent ownership.

	Vermont is creating a CIN through its RHT Program to provide shared analytics, group purchasing, administrative services, and referral formalization across its 14-hospital network.
AI scribe / ambient clinical intelligence	AI technology that listens to clinical encounters and automatically generates structured clinical documentation, reducing the documentation burden that contributes to clinician burnout. Vermont's RHT Program includes grants for AI scribe technology for rural and independent practices.
Remote patient monitoring (RPM)	Technology enabling continuous or periodic monitoring of patients' clinical parameters — heart rate, blood oxygen, blood pressure, glucose — in their homes. Enables earlier intervention for chronic conditions (CHF, COPD, diabetes) to prevent hospital admissions. Vermont's RHT Program includes RPM grants for home health, primary care, and community paramedicine settings.
42 CFR Part 2	Federal regulations governing confidentiality of substance use disorder treatment records. Separate and more restrictive than HIPAA; creates the primary legal barrier to integrating SUD treatment data into Vermont's unified health record. 2024-2025 federal regulatory changes have begun easing some Part 2 barriers.
Act 62 of 2025	Vermont legislation that transferred responsibility for the statewide Health Information Technology Plan from GMCB to DVHA, effective July 1, 2025. Aligns HIT strategy coordination with DVHA's operational relationships with providers through Medicaid, Blueprint, and AHEAD.
Master patient index	A database that matches patient identities across multiple healthcare organizations using demographic and clinical matching algorithms. Required for Vermont's integrated health record — without reliable patient matching, claims data and clinical data from different facilities cannot be reliably linked to the same individual.

Sources: GMCB VHCURES documentation (gmcboard.vermont.gov/data-and-analytics); Vermont HIE Strategic Plan 2024 Update; Act 62 of 2025; Act 68 of 2025; Oliver Wyman Act 167 Community Engagement: Recommendations (2024); Vermont Agency of Human Services / Brendan Krause, Vermont's Health Care Reform Efforts (January 31, 2025); Vermont Rural Health Transformation Program Application (November 3, 2025); Vermont RHT Program Activity Details (December 2025); Chartis 2026 State of Rural Health; JAMA Network Open ambient scribe quality improvement study (2024); National Consortium of Telehealth Resource Centers AI guidance (2026); Vermont Health Information Exchange Data Governance mandate (healthdata.vermont.gov).